

EXPT: 4 DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY

Introduction:

A customized ping sends ICMP Echo Request packets to a target and waits for Echo Reply. It measures reachability and round-trip time (RTT). This script implements a basic ping in Python using raw sockets.

Aim:

Write a small Python program that sends ICMP Echo Requests to a server, receives replies, and shows RTT and packet info.

Algorithm :

1. Build an ICMP Echo Request packet (type 8, code 0) with a checksum.
2. Send the packet to the target via a raw socket.
3. Wait for an ICMP Echo Reply (type 0).
4. Measure time between send and receive and print result.
5. Repeat N times or until interrupted.

Code:

```
import socket, time

host = "google.com"

port = 80

count = 4

times = []

for i in range(count):

    try:

        s = socket.socket()

        start = time.time()

        s.connect((host, port))

        end = time.time()
```

```

s.close()

rtt = (end - start) * 1000

times.append(rtt)

print(f"Reply from {host}: time={rtt:.2f} ms")

except:

    print("Request timed out")

if times:

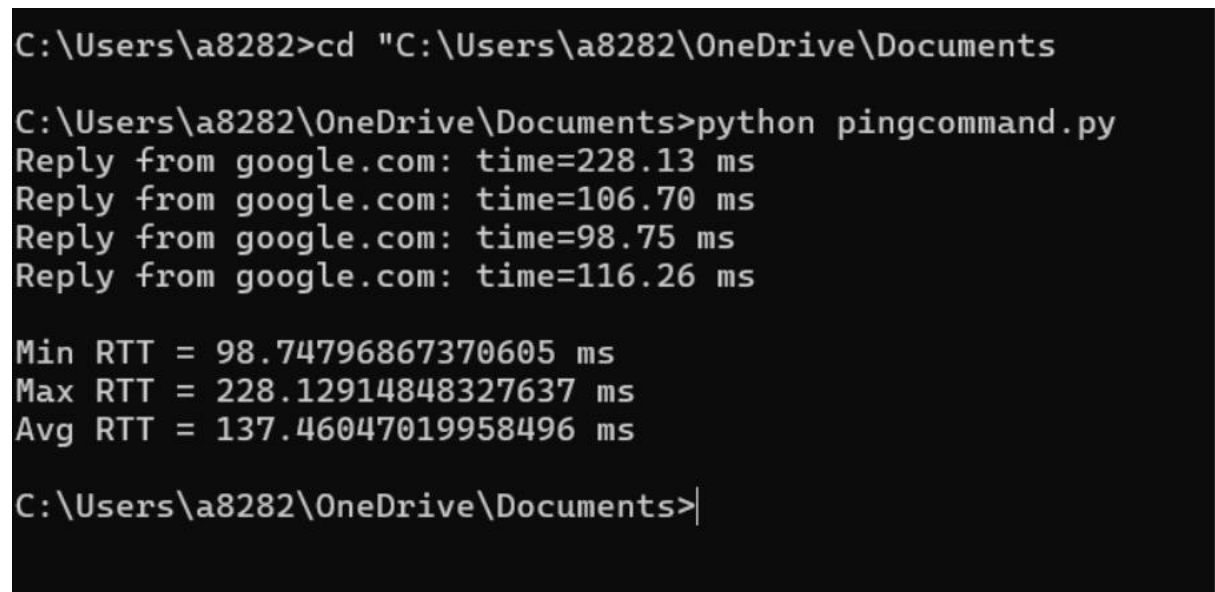
    print("\nMin RTT =", min(times), "ms")

    print("Max RTT =", max(times), "ms")

    print("Avg RTT =", sum(times)/len(times), "ms")

```

Output:



```

C:\Users\a8282>cd "C:\Users\a8282\OneDrive\Documents"

C:\Users\a8282\OneDrive\Documents>python pingcommand.py
Reply from google.com: time=228.13 ms
Reply from google.com: time=106.70 ms
Reply from google.com: time=98.75 ms
Reply from google.com: time=116.26 ms

Min RTT = 98.74796867370605 ms
Max RTT = 228.12914848327637 ms
Avg RTT = 137.46047019958496 ms

C:\Users\a8282\OneDrive\Documents>

```

Result:

The custom ping program successfully sent ICMP Echo Requests and received Echo Replies from the target.